

Karan Kakadiya

AI/ML Engineer

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SUMMARY

AI/ML Engineer with 5+ years of experience delivering **scalable, low-latency AI solutions** across cloud and edge platforms. Proven in **architecting real-time AI systems** that boost engagement, accelerate decision-making, and reduce operational bottlenecks. Skilled in **deploying and fine-tuning LLMs, RAG pipelines, and multi-agent AI frameworks**, delivering measurable gains in accuracy, speed, and adoption. Adept at **MLOps, microservices, and on-device AI**, ensuring **privacy, compliance, and production-grade reliability** in high-impact environments.

SKILLS

AI & ML: LLMs (GPT, LLaMA, Claude, Mistral, BERT), RAG, Multi-agent AI (AutoGPT, CrewAI), LoRA/PEFT fine-tuning, Federated Learning, Multi-modal AI (Text, Image, Audio), Model Compression, Prompt Engineering, ML Algorithms, MLflow, TensorFlow, PyTorch, Scikit-learn, SciPy, Hugging Face Transformers, MCP, NLP, NLTK, Embeddings

Data Engineering & MLOps: Apache Spark, Kafka, Databricks, Airflow, Vector DBs (Pinecone, FAISS, Qdrant, Chroma), SQL, NoSQL, Docker, Kubernetes, Terraform, CI/CD, AWS SageMaker, Azure ML, GCP Vertex AI

Cloud Platforms: AWS (SageMaker, Bedrock, Glue, CloudWatch, Redshift, Lambda), Azure, GCP (BigQuery, Vertex AI)

Visualization & Monitoring: Tableau, Matplotlib, Seaborn, Prometheus, Grafana, Datadog

Programming: Python, Java, R, SQL

EXPERIENCE

AI/ML Engineer | Oracle

Jan 2025 – Present

- Designed an AI-powered document processing platform using LLMs, RAG architecture, ML pipelines with vector databases and AWS services enabling real-time inference on 100K+ documents with 40% faster retrieval.
- Delivered RAG pipelines with vector databases (Qdrant, Weaviate), cutting multi-modal retrieval latency by 40%, improving customer experience in high-volume environments.
- Built API-driven AI/ML microservices (FastAPI, Flask), containerized with Docker, orchestrated via Kubernetes, and deployed across Azure & GCP, reducing deployment times by 30%.
- Implemented MLOps pipelines with MLflow, W&B, and SageMaker, improving model governance and enabling continuous experimentation, raising model accuracy by 18%.
- Developed on-device AI solutions for Android using TensorFlow Lite, MediaPipe, and PyTorch Mobile, achieving low-latency inference with federated learning, ensuring privacy compliance.
- Optimized ML models for performance through quantization, pruning, and graph compilation, reducing inference time by 35% and memory usage by 25% across edge and cloud deployments.
- Leveraged hardware acceleration (CUDA, TensorRT) for high-throughput inference on NVIDIA GPUs, improving compute efficiency by 40%.
- Implemented distributed training strategies for LLM fine-tuning and large-scale model training using PyTorch Distributed and Deep Speed, reducing training time by 50%.

ML Engineer | Accenture

Jan 2019 – July 2023

- Automated data preprocessing pipelines in Python & PySpark, cutting data prep time by 50% and improving feature quality for downstream models.
- Built predictive ML models in TensorFlow & Scikit-learn, increasing accuracy by 35% and driving higher customer conversions.
- Architected and deployed ML services with Docker, Kubernetes, and AWS/Azure, enabling real-time predictions and GPU-optimized vector search with Pinecone, FAISS, and Qdrant.
- Engineered secure cloud-native ML infrastructure with Terraform, integrated with real-time ingestion pipelines using Kafka, Databricks, and GCP BigQuery, improving throughput by 45%.
- Introduced model validation and automated testing pipelines with MLflow and pytest, ensuring 99% reproducibility and faster compliance checks.
- Optimized deep learning models (PyTorch, TensorFlow) for deployment by applying quantization-aware training, mixed-precision inference, and operator fusion, achieving 30% latency reduction.
- Enhanced inference speed on edge devices using TensorRT and ONNX Runtime, enabling 2x faster predictions for computer vision and NLP workloads.

EDUCATION

MS in Computer Science | California State University